



Graphics Cards

After processors, it's graphics cards are the next really exciting category. As there are only two major GPU chip designers – NVIDIA and AMD – it's a closely watched battle each year. While last year AMD scored over the NVIDIA cards as far as both single and dual GPU cards went, this year the tables were turned in NVIDIA's favour. In fact both GPU makers haven't announced their next series of cards as yet. Generally by this time, newer generation cards start showing up in the market, but this year we're still waiting for an official announcement on the availability of GTX 600 and HD 7000 series cards.

The GTX 500 series comprised the refreshed Fermi architecture with higher number of texture units and shader units than the GTX 400 series and it also took care of the heating and noise issues that had marred the GTX 400 series of cards. AMD started off with the HD 68xx series cards and later introduced its higher-end HD 69xx series in the single GPU segment. As time passed by, AMD and NVIDIA came out with newer cards covering every segment, and in each segment, AMD cards were outperformed by NVIDIA.

Zero1 Winner ZOTAC GTX 590

When it came to selecting one winner, we had to go with the ZOTAC GTX 590 as its scores were higher in over 50 per cent of our benchmarks. The GTX 590 comprises a pair of GF110 GPUs (i.e. GTX 580) but is clocked at a safe 607 MHz which is slower than the 772 MHz on GTX 580. The GTX 590 was better in *STALKER: Call of Pripyat*, *Resident Evil 5* and the *Unigine Heaven 2.5* benchmark whereas in *FarCry2* and *3D Mark 11* the HIS HD 6990 was better. Performance in *Crysis Warhead* was within the margin of error. Overall this card is a monster and quite great even for a three monitor setup, thanks to 3 GB of video RAM.

Worthy Mentions HIS HD 6990

This was the dual GPU card from AMD which gave a close competition to the Zero1 winner in most benchmarks. The HD 6990 is basically two Cayman GPUs in CrossFire which means two times the specs of a HD 6970 card. That means 3072 shader processors, 64 ROPs and 4GB of video RAM among other things.



ZOTAC GTX 590

Price: ₹49,000

Performance-wise it was the best as far as AMD cards went. As for the NVIDIA competition, it all boils down to what games you're going to play. Some games favour the NVIDIA card whereas AMD scores ahead in the others. So it's safe to assume that at the top end of dual GPU cards from both stables, performance really is a matter of the games you choose.



HIS HD 6990

Price: ₹39,000



ASUS GTX 580 Matrix Edition

Price: ₹35,000

ASUS GTX 580 Matrix Edition

Coming from the Republic of Gamers series, the single GPU based ASUS GTX 580 Matrix edition is one superfast card. As far as single GPU cards go, the GTX 580 outperforms the HD 6970. This card is tailor-made for overclockers thanks to its features which include Direct CU II for faster heat dissipation and the glow of the backlight on the ASUS logo on the card changes colour based on temperatures. It has ProBelt voltage readout points. The TweakIT buttons allow you to tweak the voltages on the fly and the large red button on the PCB takes the fan speed to 100 per cent. Performance-wise, like we mentioned earlier, it beats any other single GPU card from present generation. Pricing is admittedly steep but if you have the cash to spend, you won't get a more feature-rich card which performs as wonderfully.